SiMPL Electronic Batch Record — Process Instruction (PI) Sheet

AZD8630 — Affinity Chromatography (CaptureSelect CH1-XL, GPF-N, 500 L) | **NEW (XStep-based EBR)**

Material / Product	8010457 / AZD8630 Affinity (CaptureSelect CH1-XL, GPFN, 500 L)
MABR / MPR	MABR-0023643
General Supplies PO No.	____________
MPR Process Order No.	____________
Batch #	____________
Plant / Facility	GPF-North
Process	Process 1 — DSP Step 2 of 8 (Affinity Capture)
Effective Date	____________

How to read this: this is the NEW digital EBR assembled from XStep building blocks for the AZD8630 Affinity capture step. Each card shows the original MPR section it replaces ("Original BR: Section ___") so it compares directly against *MABR-0023643 / PN 8010457*. Cards are tagged NEW (built new), VARIANT (a new variant of an existing XStep), or REUSE (reuses an existing DE1 100 SMPL XStep as-is — shown as a summary, no new mock-up needed).

14 new + **24** variant + **27** reuse = **65** XSteps total.

XStep ↔ Original MPR Crosswalk

EBR Phase	XStep	Type	Original MPR Section
Process Order Information	SMPL: Order Header Details	R	Header
	SMPL: Signature Table	R	Personnel ID
	SMPL: Display BOM	R	Section 4
	SMPL: Room & Equipment Assign	R	Section 4
	SMPL: Additional Manufacturing Supplies	V	Section 4
	SMPL: Solution / Buffer Batch Record	V	Section 5
	SMPL: Additional Solution Batches Used	V	Section 5
	SMPL: Process Notes	V	Section 6
Phase 1 — AKTA Kit Install & WFI Charge	SMPL: Data Logger Verification	R	Section 7
	SMPL: Record Process Room	R	Section 7
	SMPL: WFI Container Setup & Charge	N	Section 7.4 / 7.6
	SMPL: AKTA Kit Install & Component Test	V	Section 7
Phase 2 — Column & Calculations	SMPL: Chromatography Column Information	N	Section 8.1
	SMPL: Column Pressure & Cycle Calculations	R	Section 8.2, 9
	SMPL: Column & Method Verification	R	Section 8.3
	SMPL: Load Concentration & Volume	V	Section 9.1
Phase 3 — Product Collection & Filter Setup	SMPL: Estimated VI Product Volume	N	Section 10.1
	SMPL: Product Collection Vessel Setup	N	Section 10.5
	SMPL: Product Filter Information	V	Section 10.6
	SMPL: Depth Filter Sizing Calculation	V	Section 11.1
	SMPL: Depth Filter Count Calculation	V	Section 11.2
	SMPL: Depth Filter Material Setup	R	Section 11.3

	SMPL: Total Surface Area Calculation	V	Section 11.4
	SMPL: Depth Filter Flush Flowrate Calculation	V	Section 11.5
	SMPL: Depth Filter Hold-Up Calculation	V	Section 11.6
	SMPL: Minimum WFI Flush Volume Calculation	V	Section 11.7
	SMPL: Minimum Buffer Flush Volume Calculation	V	Section 11.8
	SMPL: Record Meter ID	R	Section 11.9
	SMPL: Record Thermometer ID	R	Section 11.10
	SMPL: Pod Holder & Pressure Sensors	N	Section 11.11
	SMPL: Install Depth Filtration Units	R	Section 11.12
Phase 4 — Depth Filter Flush & Skid Connections	SMPL: Depth Filter Pre-Flush Connections	R	Section 12.1
	SMPL: Depth Filter Pre-Flush / Flush	V	Section 12.2 / 12.3
	SMPL: Skid pH Probe Standardization	V	Section 13.1
	SMPL: Skid Buffer Inlet Connections	R	Section 13.2-13.8
	SMPL: Column Guard Filter	V	Section 13.9
Phase 5 — CaptureSelect Chromatography Run	SMPL: Pre-Cycle Setup Verification	R	Section 14.1
	SMPL: POD Chase Volume Calculation	V	Section 14.5
	SMPL: Meter Standardization	R	Section 14.6-14.7
	SMPL: Chromatography Method Parameters	N	Section 14.9
	SMPL: Chromatography Method Start	N	Section 14.13-14.14
	SMPL: Phase Execution & Sign-off	V	Section 14
	SMPL: Equilibration Effluent Results	V	Section 14.15
	SMPL: Column Effluent Sample Date & Time	N	Section 14.16
	SMPL: Column Effluent Sampling	R	Section 14.17
	SMPL: Load Recording	N	Section 14.20
	SMPL: Elution Collection	N	Section 14.29-14.30
	SMPL: Strip 2 Effluent Results	V	Section 14.37-14.38
	SMPL: Strip 3 Effluent Results	V	Section 14.43-14.44
Phase 6 — Column Storage	SMPL: Column Storage Buffer & Parameters	N	Section 15.1-15.5
	SMPL: Sanitization Hold & Pre-Storage Equilibration	N	Section 14.48 / 15.6
	SMPL: Column Storage Effluent Results	V	Section 15.9
	SMPL: Column Storage Disconnect & Disposition	R	Section 15.10-15.15
Phase 7 — Product Sampling & Disposition	SMPL: Product Vessel Weigh	R	Section 16.1 / 16.7
	SMPL: Product Mixing	R	Section 16.2
	SMPL: Product Sampling & DLIMS Submission	R	Section 16.3
	SMPL: Product pH / Conductivity / Temperature	V	Section 16.4
	SMPL: Product Storage & Cross-Record	R	Section 16.8
Phase 8 — Batch Record Review & Attachments	SMPL: Batch Production Record Review	R	Review
	SMPL: Sampling Record	R	Attachment 1
	SMPL: Non-Routine Sampling Record	R	Attachment 2
	SMPL: Yield Calculations	R	Attachment 3
	SMPL: Adjusted Flowrate Table	N	Attachment 4
	SMPL: Buffer Totalizer Volumes	N	Attachment 5
	SMPL: Comments Section	R	Attachment 6

Process Order Information

SMPL: Order Header Details REUSE

Original BR: Header

Product/PN, process order numbers, recorded by — standard order header.

Reuse target: to be confirmed in DE1 100

SMPL: Signature Table REUSE

Original BR: Personnel ID

Personnel identification: print name, signature, initials.

Reuses: **SMPL Header: Signature Table**

SMPL: Display BOM REUSE

Original BR: Section 4

Bill of Materials — equipment (skid PK-4500, meters, scale, mixer) and components/filters/bags.

Reuse target: to be confirmed in DE1 100

SMPL: Room & Equipment Assign REUSE

Original BR: Section 4

Assign room + equipment (AKTA skid, pH/cond meters, thermometer, floor scale, mixer, FIT) with calibration check.

Reuses: **SMPL: Room/Equipment Assign**

SMPL: Additional Manufacturing Supplies VARIANT

Original BR: Section 4

Instructions

If additional materials (filter, bag, assembly, etc.) are used during processing, verify the material is acceptable per Section 4 and record its information; document the reason for the change in the Comments Section. Enter the Part No. — an input validation displays the Material Description. Enter the Batch No. — the Exp. Date is auto-populated. Serial No. and Autoclave Assembly ID / Exp. Date are recorded manually. **SAP consumption is posted automatically by the Goods Issue process message (Z_PICONS, movement type 261) — no manual "SAP Consumption" field is required.**

Reuses: **SMPL: Additional Assembly (with the Z_PICONS Goods Issue process message)**

Data Input

#	MPR Step No.	Part No.	Material Description	Batch No.	Exp. Date	Serial No.	Autoclave Assembly ID	Autoclave Exp. Date	Performed By *
1									

+ Add Row

Witnessed By *

SMPL: Solution / Buffer Batch Record VARIANT

Original BR: Section 5

Instructions

Record the batch number and expiration date for each process solution used in the chromatography run: Equilibration (8004949), Wash (8004950), Elution (8010534), Strips (8008812 / 8008813 / 8006409) and Column Storage (8004953). Enter the Batch No.; the Exp. Date auto-populates via input validation.

Reuses: **SMPL: Solution Summary - Data Recording**

Data Input

#	Solution / Step	Part No.	Batch No.	Exp. Date	Performed By *
1	Equilibration (50 mM Tris, pH 7.4)	8004949			

+ Add Row

Verified By *

SMPL: Additional Solution Batches Used VARIANT

Original BR: Section 5

Instructions

If a solution batch is replaced during processing, record the applicable step, part number, new batch number and expiration date. Ensure the new solution is the same part number as the one being replaced; document the change in the Comments Section. Enter the Batch Number; the Exp. Date auto-populates via input validation.

Reuses: **SMPL: Solution Summary - Data Recording**

Data Input

#	Applicable Step No.	Part Number	Batch Number	Exp. Date	Performed By *
1					

+ Add Row

Witnessed By *

SMPL: Process Notes VARIANT

Original BR: Section 6

Instructions

General process notes for the CaptureSelect CH1-XL chromatography step — review before processing:

- All activities are performed at 15 – 27 °C.
- For calculations, assume 1 L = 1 kg and 1 mL = 1 g unless otherwise noted. PK-4500 absorbance is in mAU; mAU = OD × 0.15 × 1000.
- Target linear velocity is 200 cm/hr (150 cm/hr for Elution). Alert limits ±5% of target; NOR +10%.
- Flow-rate ranges are for steady-state operation and do not consider pump ramping; temporary flowrate excursions are acceptable.
- Verify all solutions, raw materials and standards are within expiration prior to use.
- When an equipment ID is written in the MPR, the operator confirms they have visually inspected it and verified calibration prior to use.
- If PAUSE is pressed on the skid HMI the pumps stop — document the reason in the Comments Section and press CONTINUE to resume.
- During Load, "next breakpoint" may be used to proceed when the load is complete (note it in the MPR); any other use requires a supervisor/designee signature and a reason in the Comments Section. Does not apply to Elution.
- The process may be run manually if the automated chromatography recipe is not functioning correctly.
- Non-routine samples requested by MEMO are recorded in Attachment 2.
- All parameters are Targets / Alert Limits unless otherwise noted. When alert limits are exceeded, contact the Supervisor and Downstream Tech Transfer and document any actions in the Comments Section.
- To avoid over-pressurization, flow rates may be lowered manually or as part of the method; record adjusted flowrates in Attachment 4.

Reuses: **Long Text Instructions (DE2 903)**

Reviewed By * ____________

Phase 1 — AKTA Kit Install & WFI Charge

SMPL: Data Logger Verification REUSE

Original BR: Section 7

Verify the data logger is on and the PK-4500 process trend is activated (per day, first cycle).

Reuses: **Long Text Instructions (DE2 903)**

SMPL: Record Process Room REUSE

Original BR: Section 7

Record the room number where the process occurs.

Reuse target: to be confirmed in DE1 100

SMPL: WFI Container Setup & Charge NEW

Original BR: Section 7.4 / 7.6

Instructions

Set up the Day 1 / Day 2 WFI collection vessel per SOP-0080854. Inspect the collection vessel, then record the bag and WFI filter as separate lines in the table below — select the Item from the dropdown (Bag / Filter — a restricted-value CT04 characteristic), enter each Part No. and Batch No. (the Exp. Date auto-populates via input validation) and sign Performed By per line. Each line is consumed via the Goods Issue process message (Z_PICONS, mvt 261). Below the table, record the WFI valve ID, verify the WFI port is flushed and charge WFI into the vessel; stamp the charge date/time (expiration is 24 hours from charge, auto-calculated).

Data Input

#	Item	Part No.	Batch No.	Exp. Date	Performed By *
1	Bag				

+ Add Row

WFI Valve ID *

▸ Record

Date *

Time *

WFI Expiration Date & Time (Charge + 24 h)

Recorded By *

Verified By *

SMPL: AKTA Kit Install & Component Test VARIANT

Original BR: Section 7

Instructions

Obtain and install the AKTA Ready kit on skid PK-4500 per SOP-0107212. Select the Kit Required from the dropdown and enter its Part No. (the Kit Description auto-populates via input validation). Press Select Buffer to retrieve the buffer part number, then enter the Buffer Batch No. (the Exp. Date auto-populates). Run the Flow Kit Installation with Component Test (uses 5 L of 8004948 per attempt); record each attempt from the Pass / Fail / N-A dropdown and the buffer volume used. The kit and buffer are consumed via the Goods Issue process message (Z_PICONS, mvt 261).

Reuses: **SMPL: Filter Integrity Test (+ Skid ID)**

▸ Select Buffer

Buffer Part No.

Buffer Batch No. *

Buffer Exp. Date

Data Input

#	Kit Required	Part No.	Kit Description	Test Program	Test 1	Test 2 (i/a)	Test 3 (i/a)	Buffer Volume Used (L)	Skid ID	Performed By *
1	AKTA High Flow Kit								PK-4500	

+ Add Row

Witness By *

Phase 2 — Column & Calculations

SMPL: Chromatography Column Information NEW

Original BR: Section 8.1

Instructions

Obtain the CaptureSelect CH1-XL column and the Column Packing Data Sheet (CPDS, FORM-0064069). Verify the column was packed with CaptureSelect CH1-XL for AZD8630. Record the Column ID, media name, CPDS lot/part number, bed volume, inlet/differential pressure and bed height from the CPDS (all entered manually from the CPDS). Assign the Column Pressure Rating as the Inlet Pressure and the Packing Pressure as the Differential Pressure. The column is consumed via the Goods Issue process message (Z_PICONS, mvt 261).

Column ID *

Chromatography Media Name *

CPDS Lot No. *

CPDS Part Number *

Column Bed Volume (L) *

Column Inlet Pressure (bar) *

Column Differential Pressure (barD) *

Column Bed Height (16-22 cm) *

Performed By *

Witnessed By *

SMPL: Column Pressure & Cycle Calculations REUSE

Original BR: Section 8.2, 9

Calc chain: inlet pressure max (x0.9), total protein, max load (15 g/L/cycle), number of cycles, load volume per cycle.

Reuses: **SMPL: Three Variable Calc (calc chain)**

SMPL: Column & Method Verification REUSE

Original BR: Section 8.3

Verify from the column logbook it was sanitized/stored correctly and the AZD8630_ProteinA method was wet-tested & ran as expected.

Reuses: **Long Text Instructions (DE2 903)**

SMPL: Load Concentration & Volume VARIANT

Original BR: Section 9.1

Instructions

Obtain the protein concentration and net volume of the Conditioned Medium (CM, PN 8010441) for each bag from the AZD8630 Pod Harvest MPR (MABR-0023642) and record them below with the DLIMS project and sample numbers (one row per CM bag; values carry forward from Pod Harvest). The CM is consumed via the Goods Issue process message (Z_PICONS, mvt 261).

Reuses: **SMPL: Solution Summary - Data Recording**

Data Input

Bag No.	CM Product Net Volume (kg = L)	CM Concentration (g/L)	DLIMS Project Number	DLIMS Sample Number	Performed By *
1					

+ Add Row

Witness By *

Phase 3 — Product Collection & Filter Setup

SMPL: Estimated VI Product Volume NEW

Original BR: Section 10.1

Instructions

Calculate the estimated Viral Inactivation (VI) product volume — a 4-variable calculation (reuses the Three Variable Calc layout, extended to four variables). Column Bed Volume carries from the Column Information step and Number of Cycles from the Chromatography Calculations; enter 2.3 CV/cycle for Estimated Product Volume; Estimated VI Additions is the standard 1.10. The result sizes the product collection vessel.

Reuses: **new SMPL: 4 Variable Calc (based on SMPL: Three Variable Calc)**

Data Input

Column Bed Volume (L/CV)		Estimated Product Volume (CV/cycle)		Number of Cycles		Estimated VI Additions		Estimated VI Product Volume (L)	Performed By *
	×		×		×	1.10	=		

Witnessed By *

SMPL: Product Collection Vessel Setup NEW

Original BR: Section 10.5

Instructions

Obtain an appropriately sized product vessel based on the estimated VI product volume. Press Get Equipment to select the scale/balance — its ID and Calibration Due Date are retrieved. Record the tare weight and the collection bag information (enter the Batch No.; the Exp. Date auto-populates). Indicate which attachments are connected to the LevTech bag during the vessel tare (Mixer Drive / Top Base / Magnetic Clamp; N/A if no LevTech bag). Label the vessel "AZD8630", "CaptureSelect CH1-XL Product". The collection bag is consumed via the Goods Issue process message (Z_PICONS, mvt 261).

▸ Get Equipment

Scale / Balance ID

Calibration Due Date

Tare Weight of Product Vessel (kg) *

Collection Bag Part No. *

Collection Bag Batch No. *

Collection Bag Exp. Date

Mixer Drive Attached? *

Top Base Attached? *

Magnetic Clamp Attached? *

Recorded By *

Witnessed By *

SMPL: Product Filter Information VARIANT

Original BR: Section 10.6

Instructions

Record the product filter(s) attached to the collection vessel (one line per filter, Day 1 / Day 2). Enter the Part No. — an input validation displays the Material Description. Enter the Batch No. — the Exp. Date is auto-populated. Serial No. and Autoclave Assembly ID / Exp. Date are recorded manually. SAP consumption is posted automatically by the Goods Issue process message (Z_PICONS, movement type 261).

Reuses: **SMPL: Additional Assembly (same as Additional Manufacturing Supplies, without MPR Step No.)**

Data Input

Day #	Part No.	Material Description	Batch No.	Exp. Date	Serial No.	Autoclave Assembly ID	Autoclave Exp. Date	Performed By *
1								

+ Add Row

Witnessed By *

SMPL: Depth Filter Sizing Calculation VARIANT

Original BR: Section 11.1

Instructions

Calculate the C0HC depth-filter surface area needed for the day — reuses the Three Variable Calc XStep, run as one instance per day (Day 1 uses Bag 1 volume, Day 2 uses Bag 2): CM Volume per Day ÷ C0HC Capacity (500 L/m²) = C0HC Surface Area Needed.

Reuses: **SMPL: Three Variable Calc (one instance per day)**

Data Input

CM Volume per Day (L)		C0HC Capacity (L/m²)		C0HC Surface Area Needed (m²)	Performed By *
	÷	500	=		

Witnessed By *

SMPL: Depth Filter Count Calculation VARIANT

Original BR: Section 11.2

Instructions

Calculate the number of C0HC depth-filter units to install for the day (round up to the next whole number) — reuses the Three Variable Calc XStep, one instance per day: C0HC Surface Area Needed (from the Depth Filter Sizing Calculation) ÷ C0HC Filter Area (1.1 m²) = Number of C0HC Filters.

Reuses: **SMPL: Three Variable Calc (one instance per day)**

Data Input

C0HC Surface Area Needed (m²)		C0HC Filter Area (m²)		Number of C0HC Filters (round up)	Performed By *
	÷	1.1	=		

Witnessed By *

SMPL: Depth Filter Material Setup REUSE

Original BR: Section 11.3

Record C0HC filters and POD adaptor kits used (Description/Qty/Batch/Exp/Serial + SAP consumption).

Reuses: **SMPL: Material Consumption**

SMPL: Total Surface Area Calculation VARIANT

Original BR: Section 11.4

Instructions

Calculate the total surface area of the C0HC units — reuses the Three Variable Calc XStep, one instance per day: Number of C0HC Filters × Surface Area per Unit (1.1 m²) = Total Surface Area of C0HC Units.

Reuses: **SMPL: Three Variable Calc (one instance per day)**

Data Input

Number of C0HC Filters		Surface Area per Unit (m²)		Total Surface Area of C0HC Units (m²)	Performed By *
	×	1.1	=		

Witnessed By *

SMPL: Depth Filter Flush Flowrate Calculation VARIANT

Original BR: Section 11.5

Instructions

Calculate the flowrate range to flush the depth filtration units — reuses the Three Variable Calc XStep, one instance per day: Flux × Total Surface Area of C0HC Units (Step 11.4) = Flow Rate Range. Flux is 500 (min) / 600 (target) / 700 (max) LMH (NOR allows below 500 LMH if the equipment cannot meet the minimum).

Reuses: **SMPL: Three Variable Calc (one instance per day)**

Data Input

Flux (LMH)		Total Surface Area of C0HC Units (m²)		Flow Rate Range (L/hr)	Performed By *
500 / 600 / 700	×		=		

Witnessed By *

SMPL: Depth Filter Hold-Up Calculation VARIANT

Original BR: Section 11.6

Instructions

Calculate the hold-up volume of the installed depth filters — reuses the Three Variable Calc XStep, one instance per day: Number of C0HC Filters × Internal Void Volume per 1.1 m² Filter (10.3 L) = Hold-Up Volume of Installed Depth Filters.

Reuses: **SMPL: Three Variable Calc (one instance per day)**

Data Input

Number of C0HC Filters		Internal Void Volume per Filter (L)		Hold-Up Volume of Installed Filters (L)	Performed By *
	×	10.3	=		

Witnessed By *

SMPL: Minimum WFI Flush Volume Calculation VARIANT

Original BR: Section 11.7

Instructions

Calculate the minimum volume of WFI to flush the depth filtration units — reuses the Three Variable Calc XStep, one instance per day: WFI Min (105 L/m²) × Total Surface Area of C0HC Units (Step 11.4) = Minimum WFI Flush Volume.

Reuses: **SMPL: Three Variable Calc (one instance per day)**

Data Input

WFI Min (L/m²)		Total Surface Area of C0HC Units (m²)		Minimum WFI Flush Volume (L)	Performed By *
105	×		=		

Witnessed By *

SMPL: Minimum Buffer Flush Volume Calculation VARIANT

Original BR: Section 11.8

Instructions

Calculate the minimum volume of Equilibration Buffer (8004949: 50 mM Tris, pH 7.4) to flush the depth filtration units — reuses the new 4 Variable Calc XStep (mixed × and + operators), one instance per day: Buffer Min (30 L/m²) × Total Surface Area of C0HC Units (Step 11.4) + Hold-Up Volume of Installed Depth Filters (Step 11.6) = Minimum Buffer Flush Volume.

Reuses: **new SMPL: 4 Variable Calc (based on SMPL: Three Variable Calc), one instance per day**

Data Input

Buffer Min (L/m²)		Total Surface Area of C0HC Units (m²)		Hold-Up Volume of Installed Filters (L)		Minimum Buffer Flush Volume (L)	Performed By *
30	×		+		=		

Witnessed By *

SMPL: Record Meter ID REUSE

Original BR: Section 11.9

Confirm the pH/conductivity meter is standardized per SOP-0080297; record the Meter ID (Day 1 only). Reuses SMPL: Record Text Value.

Reuses: **SMPL: Record Text Value (DE1 100)**

SMPL: Record Thermometer ID REUSE

Original BR: Section 11.10

Confirm the thermometer is within its calibration interval; record the Thermometer ID (Day 1 only). Reuses SMPL: Record Text Value.

Reuses: **SMPL: Record Text Value (DE1 100)**

SMPL: Pod Holder & Pressure Sensors NEW

Original BR: Section 11.11

Instructions

Obtain and document the depth-filter Pod Holder ID and the inlet/outlet pressure sensors used (Day 1 / Day 2). Record the Pod Holder ID above the table; for each pressure sensor select Inlet or Outlet and record its Part No. and Batch No. (Exp. Date auto-populates), then sign Performed By — signing a line triggers the Goods Issue (Z_PICONS, mvt 261) consumption of that sensor. Press Get Pendotech to retrieve the Pendotech ID and its Calibration Due Date.

Pod Holder ID *

Data Input

#	Inlet / Outlet	Part No.	Batch No.	Exp. Date	Performed By *
1	Inlet				

+ Add Row

▸ Get Pendotech

Pendotech ID

Pendotech Calibration Due Date

Witnessed By *

SMPL: Install Depth Filtration Units REUSE

Original BR: Section 11.12

Install the depth filtration units into the skid column position per SOP-0107204 (POD inlet/outlet T's, pressure-sensor connections, bypass line, pinch-clamp flow) and sign off (Day 1 / Day 2). Reuses Long Text Instructions.

Reuses: **Long Text Instructions (DE2 903)**

Phase 4 — Depth Filter Flush & Skid Connections

SMPL: Depth Filter Pre-Flush Connections REUSE

Original BR: Section 12.1

Make the depth-filter pre-flush connections per the recipe (Day 1 / Day 2): Inlet 2 to the Equilibration buffer (8004949: 50 mM Tris, pH 7.4), Inlet 1 to WFI, and Outlet 1 to Waste. Sign off. Reuses Long Text Instructions.

Reuses: **Long Text Instructions (DE2 903)**

SMPL: Depth Filter Pre-Flush / Flush VARIANT

Original BR: Section 12.2 / 12.3

Instructions

With the pre-flush connections made per Step 12.1 (Inlet 2: 8004949 50 mM Tris; Inlet 1: WFI; Outlet 1: Waste), flush the depth-filter pods per SOP-0080429 — bleed the vent valves first. For each flush, perform a Manual Run on the AKTA skid (flow paths method-driven; Alarm PT111 High 3.4 bar, Delta 1.3 bar; flow 25% until bled, then target per Step 11.5) labelled "Batch ID: PartNumber BatchNumber DepthFilterFlush Date", and record the Batch ID and the POD flush volume: the WFI Flush must be ≥ Step 11.7 and the Equilibration Flush ≥ Step 11.8. Adjust flow to stay under the pressure alarms. The Equilibration buffer flush is consumed via the Goods Issue process message (Z_PICONS, mvt 261). One instance per processing day.

Reuses: **SMPL: Additional Assembly (Equil Flush line carries the Z_PICONS Goods Issue)**

Data Input

Day #	Flush Step	Batch ID	Flush Volume (L)	Performed By *
1	Depth Filter WFI Flush			

+ Add Row

Witnessed By *

SMPL: Skid pH Probe Standardization VARIANT

Original BR: Section 13.1

Instructions

Verify the chromatography skid pH probe has been user-standardized per SOP-0107212 and record the probe information. Enter the Batch No. — the Exp. Date auto-populates via input validation. The pH probe (6003953) is consumed via the Goods Issue process message (Z_PICONS, mvt 261). One instance per processing day.

Reuses: **SMPL: Additional Assembly (with the Z_PICONS Goods Issue process message)**

Data Input

Day #	Part No.	Material Description	Batch No.	Exp. Date	Performed By *
1	6003953				

+ Add Row

Witnessed By *

SMPL: Skid Buffer Inlet Connections REUSE

Original BR: Section 13.2-13.8

Connect the process buffers to the chromatography-skid inlets per SOP-0107212 (Inlet 1: Conditioned Medium 8010441 Load; Inlet 2: 8004949 Equilibration 1-5; Inlet 3: 8004950 Wash; Inlet 4: 8010534 Elution; Inlet 5: 8008812 Strip 1; Inlet 6 via Y-connector: 8008813 Strip 2 / 8006409 Strip 3). Set the Inlet 6 Y-leg clamps (8008813 Open, 8006409 Closed) and, for a 1-day 2-bag process, connect both CM bags to Inlet 1 with a Y. Day 1 / Day 2 sign-off. Reuses Long Text Instructions.

Reuses: **Long Text Instructions (DE2 903)**

SMPL: Column Guard Filter VARIANT

Original BR: Section 13.9

Instructions

Obtain and install the column guard filter (preferred: 30″ SHC filter 4100987; does not need to be autoclaved) — inlet to the POD outlet T, outlet to the column tubing inlet; adjust both T pinch-clamps to bypass the POD(s). Enter the Part No. (Material Description auto-populates) and Batch No. (Exp. Date auto-populates), and record the Serial No. The guard filter is consumed via the Goods Issue process message (Z_PICONS, mvt 261). One instance per processing day.

Reuses: **SMPL: Additional Assembly (with the Z_PICONS Goods Issue process message)**

Data Input

Day #	Part No.	Material Description	Batch No.	Exp. Date	Serial No.	Performed By *
1	4100987					

+ Add Row

Witnessed By *

Phase 5 — CaptureSelect Chromatography Run

SMPL: Pre-Cycle Setup Verification REUSE

Original BR: Section 14.1

Verify CM held ≥12 h at 2-8 °C; column connected; guard-filter vent open, column inlet closed, bleed open, outlet open; product vessel connected (per cycle).

Reuses: **Long Text Instructions (DE2 903)**

SMPL: POD Chase Volume Calculation VARIANT

Original BR: Section 14.5

Instructions

Calculate the POD chase volume — reuses the SMPL: Calc Three Columns XStep (Value 1 × Value 2 = Result), run as one instance per processing day: Number of C0HC Filters (Step 11.3) × POD Hold-Up Volume (10.6 L) = POD Chase Volume.

Reuses: **SMPL: Calc Three Columns**

Data Input

Number of C0HC Filters		POD Hold-Up Volume (L)		POD Chase Volume (L)	Performed By *
	×	10.6	=		

Witnessed By *

SMPL: Meter Standardization REUSE

Original BR: Section 14.6-14.7

Confirm the thermometer is within its calibration interval and record the Thermometer ID; confirm the pH meter is standardized between pH 4.0 and 10.0 and the conductivity meter is standardized per SOP-0080297, and record the Meter ID. Reuses SMPL: Record Text Value.

Reuses: **SMPL: Record Text Value (DE1 100)**

SMPL: Chromatography Method Parameters NEW

Original BR: Section 14.9

Instructions

Initiate the AZD8630_ProteinA method per SOP-0107212, document the variables below and input them into the method. Bed Volume and Differential Pressure carry from Step 8.1, Inlet Pressure Max from Step 8.2, and Load Volume from Step 9.13 (Day 1 of 2) / 9.16 (Day 2 of 2) / 9.10 (both bags in one day). If a CPDS value exceeds the skid rating, record the skid maximum instead (5 bar / 4 barD).

AZD8630_ProteinA Method Overview

Volumes	Target Flowrate	Step
3 CV	318 L/hr	Equilibration 1
Step 9.13 / 9.16 / 9.10	318 L/hr	Load
3 CV	318 L/hr	Equilibration 2
3 CV	318 L/hr	Wash
3 CV	318 L/hr	Equilibration 3
N/A	239 L/hr	Elution — Collection Start: 37.5 mAU / Collection End: 37.5 mAU
3 CV	318 L/hr	Strip 1
1 CV	318 L/hr	Equilibration 4
3 CV	318 L/hr	Strip 2
1 CV	318 L/hr	Equilibration 5
3 CV	318 L/hr	Strip 3

Data Input

Cycle #	Bed Volume (L)	Inlet Press. Max (bar)	Diff. Press. (barD)	Load Volume (L)	Performed By *
1					

+ Add Row

Verified By *

SMPL: Chromatography Method Start NEW

Original BR: Section 14.13-14.14

Instructions

Start the AZD8630_ProteinA method per SOP-0107212. For each cycle, enter the batch Name ID (Format: PartNumber_ProductBatchNumber_Cycle#_AZD8630_ProteinA_Date) and stamp the start time. Allowed during the run: Flowrate 302-334 L/hr (Elution 226-251 L/hr), NOR 0.0-350 L/hr; Process CV -0.1 to +1.0 CV from target. Verify the flow paths match the recipe — contact area management immediately on any discrepancy. Record any lowered flowrate in Attachment 4.

Data Input

Cycle #	Cycle Name ID		Start Date	Start Time	Performed By *
1		▸ Record			

+ Add Row

Witnessed By *

SMPL: Phase Execution & Sign-off VARIANT

Original BR: Section 14

Instructions

Execute the AZD8630_ProteinA method phases per cycle and sign off. Method sequence (Inlet / Flow / Volume are method-driven):

- Skid Flush, Column Guard Filter Purge, Column Bleed Line Purge (8004949).
- Equilibration 1 (3 CV) → Load (POD inline) → Equilibration 2 (3 CV).
- Wash (8004950, 3 CV) → Equilibration 3 (3 CV).
- Elution (8010534, collect 37.5→37.5 mAU) → Strip 1 (8008812) → Equilibration 4.
- Strip 2 (8008813) → Equilibration 5 → Strip 3 (8006409).

Reposition POD clamps inline/bypass when the skid pauses; press CONTINUE. Record any PAUSE / flow-rate change in the Comments / Adjusted Flowrate attachments.

Reuses: **Long Text Instructions (DE2 903)**

Performed By *

Witnessed By *

SMPL: Equilibration Effluent Results VARIANT

Original BR: Section 14.15

Instructions

Confirm the equilibration effluent is in range using an offline meter and record the pH, conductivity and temperature for each cycle (after 2 CVs of Equilibration 1). Get the offline meter — the Equipment ID and Description auto-populate. Verify the Offline pH is within 7.2 - 7.6 (NOR 7.0 - 7.8) and the Temperature within 15 - 25 °C; record the CV at which each reading was taken. **Contact the area supervisor/designee if a result is out of specification.** One row per cycle; add rows as needed.

Reuses: **SMPL: Solution Summary - Data Recording (range-validated variant)**

▸ Get Equipment

Equipment ID

Equipment Description

Data Input

Cycle #	Offline pH	UoM	Conductivity	UoM	Temperature	UoM	CV Where Taken	Performed By *
1		pH		mS/cm		°C		

+ Add Row

Witnessed By *

SMPL: Column Effluent Sample Date & Time NEW

Original BR: Section 14.16

Instructions

Record the date and time the Affinity Chromatography Column Effluent sample is removed, per processing day (first cycle of each day only). Press Get Date/Time to stamp the current Date and Time, then sign Performed By. Add one row per processing day.

Data Input

Day #		Date	Time	Performed By *
1	▸ Get Date/Time			

+ Add Row

Witnessed By *

SMPL: Column Effluent Sampling REUSE

Original BR: Section 14.17

First cycle of each day: sample Affinity Column Effluent (8010457-20-1 Day 1 / -20-2 Day 2) per Attachment 1, stamp date/time, aliquot, label, submit to DLIMS.

Reuses: **SMPL: Sampling Record**

SMPL: Load Recording NEW

Original BR: Section 14.20

Instructions

Load the Conditioned Medium (8010441) onto the column (Inlet 1, 318 L/hr). For each cycle, stamp the Load start and end times and record the Actual Volume Loaded from the skid totalizer. The total load volume is summed across all cycles.

Data Input

Cycle #	Start	Load Start Time	End	Load End Time	Vol Loaded /Totalizer (L)	Performed By *
1	▸ Start		▸ End			

+ Add Row

Total Load Volume, All Cycles (L)

Witnessed By *

SMPL: Elution Collection NEW

Original BR: Section 14.29-14.30

Instructions

Collect the elution peak into the product vessel (8010534: 25 mM Sodium Acetate, pH 3.5, 239 L/hr). Collection begins when the absorbance reaches 37.5 mAU and the outlet switches to Outlet 4; bleed air from the product filter at the start. When the UV returns to 37.5 mAU the outlet switches back to Outlet 1. Record the product net weight / volume from the Outlet 4 totalizer (or weigh the eluate if the totalizer is unavailable) and stamp the Elution End Time, per cycle.

Data Input

Cycle #	Net Wt / Vol (kg=L)	End	Elution End Time	Performed By *
1		▸ End	⏱	

+ Add Row

Witnessed By *

SMPL: Strip 2 Effluent Results VARIANT

Original BR: Section 14.37-14.38

Instructions

Per cycle, verify the Strip 2 effluent conductivity is within the NOR of 2.2 - 3.8 mS/cm and the temperature is in range 15 - 27 °C using an offline meter (Get Equipment — the Equipment ID and Description auto-populate). Also record the Inlet 6 totalizer volume at Strip 2 completion (transcribed to Attachment 5: Buffer Totalizer Volumes). **Contact a supervisor if the conductivity is out of specification.** One row per cycle.

Reuses: **SMPL: Solution Summary - Data Recording (range-validated variant)**

▸ Get Equipment

Equipment ID

Equipment Description

Data Input

Cycle #	Conductivity	UoM	Temperature	UoM	Inlet 6 Totalizer Vol (L)	Performed By *
1		mS/cm		°C		

+ Add Row

Witnessed By *

SMPL: Strip 3 Effluent Results VARIANT

Original BR: Section 14.43-14.44

Instructions

Confirm the pH meter has been standardized between pH 2.0 and 7.0 per SOP-0080297 (Get Equipment — the Equipment ID/Description auto-populate). Per cycle, verify the Strip 3 effluent pH is ≤ 2.4 (NOR ≤ 2.5) and the temperature is in range 15 - 27 °C using an offline meter. **Contact a supervisor if the pH is out of specification.** One row per cycle.

Reuses: **SMPL: Solution Summary - Data Recording (range-validated variant)**

▸ Get Equipment

Equipment ID

Equipment Description

Data Input

Cycle #	Offline pH	UoM	Temperature	UoM	Performed By *
1		pH		°C	

+ Add Row

Witnessed By *

Phase 6 — Column Storage

SMPL: Column Storage Buffer & Parameters NEW

Original BR: Section 15.1-15.5

Instructions

Record the Column Storage buffer (8004953: 2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0) — enter the Batch No. (the Exp. Date auto-populates via input validation) — and connect it to Inlet 6. Initiate the AZD8630 Affinity Storage method per SOP-0107212 and verify the method parameters carried from Sections 8 and 9 (bed volume, inlet pressure max, differential pressure; flow 318 L/hr, NOR 0.0-350 L/hr). Record the Name ID and stamp the storage Start Date & Time. The storage buffer is consumed via the Goods Issue process message (Z_PICONS, mvt 261). One instance per processing day.

Storage Buffer PN (8004953)	
Storage Buffer Batch No. *	
Storage Buffer Exp. Date	
Column Bed Volume (L)	
Inlet Press. Max (bar)	
Diff. Pressure (barD)	
Storage Flow Rate (L/hr) *	
Storage Method Name ID *	
	▸ Record
Date *	
Time *	
Performed By *	
Verified By *	

SMPL: Sanitization Hold & Pre-Storage Equilibration NEW

Original BR: Section 14.48 / 15.6

Instructions

After Strip 3 of the final cycle, hold the column for column sanitization with 8006409 (120 mM phosphoric acid, 167 mM acetic acid, 2.2% benzyl alcohol), NOR 30-120 min (target 35). The Hold End Time is when the pump starts for Pre-Storage Equilibration (8004949: 50 mM Tris, pH 7.4). Stamp the Sanitization Hold Start Time (Step 14.48) and the Hold End Time; the Hold Duration is calculated (End − Start). One instance per processing day.

	▸ Record
Sanitization Hold Start Time *	
	▸ Record
Hold End Time *	
Hold Duration (min)	
Performed By *	
Witnessed By *	

SMPL: Column Storage Effluent Results VARIANT

Original BR: Section 15.9

Instructions

At the end of Column Storage, verify the effluent is in range using an offline meter (Get Equipment — the Equipment ID and Description auto-populate) and record the pH and temperature per processing day. Verify the pH is within 4.8 - 5.2 and the temperature within 15 - 25 °C. **Contact the area supervisor/designee if the pH is out of specification.** One row per day.

Reuses: **SMPL: Solution Summary - Data Recording (range-validated variant)**

▸ Get Equipment

Equipment ID

Equipment Description

Data Input

Day #	Offline pH	UoM	Temperature	UoM	Performed By *
1		pH		°C	

+ Add Row

Witnessed By *

SMPL: Column Storage Disconnect & Disposition REUSE

Original BR: Section 15.10-15.15

After the Storage step completes, store the pH probe in KCl storage solution (end of Day 1, if applicable) and record the buffer totalizer volumes in Attachment 5. Disconnect the column and dispose of the column guard filter; disconnect/discard the load tubing (and the AKTA Ready Kit if processing is complete); close the product-vessel inlet, disconnect it from Outlet 4 and dispose of the product filter. For a 2-day process, record the Day 1 product interim storage start time/date and temperature (2-8 / 15-25 °C). Reuses Long Text Instructions.

Reuses: **Long Text Instructions (DE2 903)**

Phase 7 — Product Sampling & Disposition

SMPL: Product Vessel Weigh REUSE

Original BR: Section 16.1 / 16.7

Weigh the product vessel (scale ID + calibration); gross & net weight (=gross − tare) before and after sampling.

Reuses: **SMPL: Bag Replacement Weight (weight pattern)**

SMPL: Product Mixing REUSE

Original BR: Section 16.2

Mix product ≥15 min (LevTech/Mobius): mixer ID, agitation RPM, start/end time, mixing duration.

Reuses: **SMPL: Bag Replacement Weight / weight+timer pattern**

SMPL: Product Sampling & DLIMS Submission REUSE

Original BR: Section 16.3

Remove product sample (8009938-30) per Attachment 1, stamp date/time, aliquot, label, submit to DLIMS. Reuses SMPL: Sampling Record.

Reuses: **SMPL: Sampling Record**

SMPL: Product pH / Conductivity / Temperature VARIANT

Original BR: Section 16.4

Instructions

Place 10 mL of product in a 50 mL conical tube and measure the pH, conductivity and temperature using an offline meter (Get Equipment — the Equipment ID and Description auto-populate). Record the results on the final processing day.

Reuses: **SMPL: Solution Summary - Data Recording (range-validated variant)**

▸ Get Equipment

Equipment ID

Equipment Description

Data Input

#	Offline pH	UoM	Conductivity	UoM	Temperature	UoM	Performed By *
1		pH		mS/cm		°C	

+ Add Row

Witnessed By *

SMPL: Product Storage & Cross-Record REUSE

Original BR: Section 16.8

Record initial storage condition (2-8 / 15-25 °C) + storage start time/date; cross-write Load Start Time back to the Pod Harvest MPR.

Reuse target: to be confirmed in DE1 100

Phase 8 — Batch Record Review & Attachments

SMPL: Batch Production Record Review REUSE

Original BR: Review

Verify supporting docs attached, net weight in SAP, yield calcs complete, all materials/CPDS consumed into SAP; submit for review.

Reuse target: to be confirmed in DE1 100

SMPL: Sampling Record REUSE

Original BR: Attachment 1

DLIMS sample submission chart (GREENFORM): -10 starting / -20 column effluent / -30 product; test names, volumes, tube types, storage temps, sample numbers. Reuses SMPL: Sampling Record / Sample Submission Chart.

Reuses: **SMPL: Sampling Record + SMPL: Sample Submission Chart**

SMPL: Non-Routine Sampling Record REUSE

Original BR: Attachment 2

Ad-hoc samples requested by MEMO: process section, contents, container, quantity, designation, DLIMS. Reuses SMPL: Non-Routine Sampling Record.

Reuses: **SMPL: Non-Routine Sampling Record**

SMPL: Yield Calculations REUSE

Original BR: Attachment 3

Load mass (pre-sampling), final product mass (pre/post-sampling), % yield. Reuses calculation XSteps.

Reuses: **SMPL: Yield Calculations**

SMPL: Adjusted Flowrate Table NEW

Original BR: Attachment 4

Instructions

If flow rates are lowered to avoid over-pressurization during the run, record the step number, cycle number, adjusted flow rate achieved, and the reason for the change.

Data Input

#	Step Number	Cycle Number	Adjusted Flow Rate (L/hr)	Reason for Flow Rate Change	Performed By *
1					

+ Add Row

Witnessed By *

SMPL: Buffer Totalizer Volumes NEW

Original BR: Attachment 5

Instructions

Record the buffer totalizer volume consumed per cycle for each buffer/inlet, plus the total. Buffers: Tris (8004949), Tris/NaCl wash (8004950), Elution (8010534), Strips (8008812 / 8008813 / 8006409) and Storage (8004953). Buffer consumption is posted via the Goods Issue process message (Z_PICONS, mvt 261).

Data Input

Inlet	Buffer	Cycle 1 (L)	Cycle 2 (L)	Cycle 3 (L)	Cycle 4 (L)	Cycle 5 (L)	Total (L)	Performed By *
2	50 mM Tris, pH 7.4 (8004949)							

+ Add Row

Verified By *

SMPL: Comments Section REUSE

Original BR: Attachment 6

Record comments, deviations, PAUSE reasons, and corrective actions (step no. + initials/date). Reuses SMPL: Comments.

Reuse target: to be confirmed in DE1 100